

Classifier Combination Supported by the Sleep-Wake Cycle Improves EEG Seizure Prediction Performance

Ana Oliveira , Mauro F. Pinto , Fábio Lopes , Adriana Leal , and César A. Teixeira 

Abstract—Objective: Seizure prediction is a promising solution to improve the quality of life for drug-resistant patients, which concerns nearly 30% of patients with epilepsy. The present study aimed to ascertain the impact of incorporating sleep-wake information in seizure prediction. **Methods:** We developed five patient-specific prediction approaches that use vigilance state information differently: i) using it as an input feature, ii) building a pool of two classifiers, each with different weights to sleep/wake training samples, iii) building a pool of two classifiers, each with only sleep/wake samples, iv) changing the alarm-threshold concerning each sleep/wake state, and v) adjusting the alarm-threshold after a sleep-wake transition. We compared these approaches with a control method that did not integrate sleep-wake information. Our models were tested with data (43 seizures and 482 hours) acquired during presurgical monitoring of 17 patients from the EPILEPSIAE database. As EPILEPSIAE does not contain vigilance state annotations, we developed a sleep-wake classifier using 33 patients diagnosed with nocturnal frontal lobe epilepsy from the CAP Sleep database. **Results:** Although different patients may require different strategies, our best approach, the pool of weighted predictors, obtained 65% of patients performing above chance level with a surrogate analysis (against 41% in the control method). **Conclusion:** The inclusion of vigilance state information improves seizure prediction. Higher results and testing with long-term recordings from daily-life conditions are necessary to ensure clinical acceptance. **Significance:** As automated

sleep-wake detection is possible, it would be feasible to incorporate these algorithms into future devices for seizure prediction.

Index Terms—Drug-resistant epilepsy, electroencephalogram, seizure prediction, sleep-wake cycle.

I. INTRODUCTION

EPILEPSY is a neurological disease that affects around 1% of the global population and is characterized by recurrent epileptic seizures, which may cause loss of awareness and motor control [1]. Despite seizures usually occupying less than 0.1% of the time [2], their sudden occurrence brings consequences at many levels: physical, psychological, and social. Approximately 70% of patients achieve seizure-free lives with antiepileptic drugs [3]. For Drug-Resistant Epilepsy (DRE) patients, surgery is a potential treatment, albeit only effective in certain cases. Therefore, predicting seizures can be crucial as it allows the development of closed-loop intervention systems or, at least, warning systems to take precautions for patients' safety.

The main purpose of seizure prediction is to develop an algorithm capable of anticipating seizures and triggering alarms according to changes in brain dynamics. This field assumes that Electroencephalogram (EEG) signals are the best indicators of these changes [4]. While the ideal scenario would be to predict the exact seizure onset time, some uncertainty is presumably unavoidable. Accordingly, an alarm is associated with a defined time interval for taking action, called Seizure Prediction Horizon (SPH), and a period of time during which the seizure is expected to happen, known as Seizure Occurrence Period (SOP) [5].

Over the years, there has been increasing evidence about the existence of a preictal state, which is the period that represents the transition between a normal state and a seizure [6]. However, its establishment poses a challenge and a significant limitation since there is no recurrent pattern, and its heterogeneity spans not only within patients but also between seizures from the same patient [6], [7], [8]. As seizures are relatively rare events, addressing data imbalance and limited data availability of these events are other important issues to consider [6].

Also, factors like medication, circadian rhythms, and stress, among others, may influence the brain dynamics and consequently affect the susceptibility to seizures [4], [8], [9], [10]. These factors can also confound EEG measures, which can

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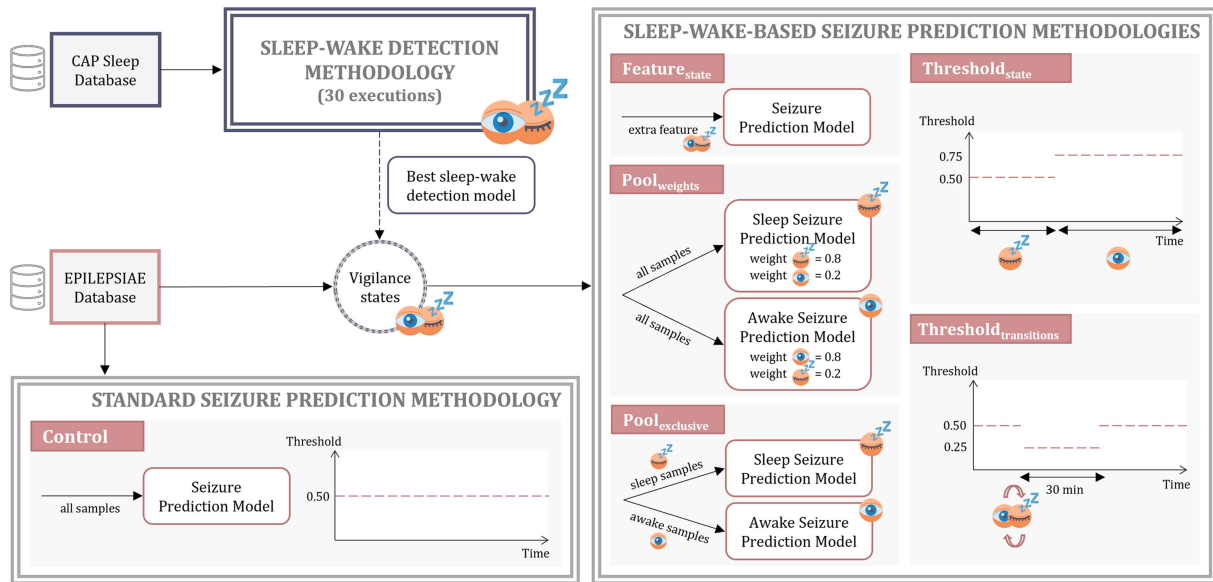


Fig. 1. Schematic representation of the proposed pipeline, comprising the sleep-wake detection model, the standard seizure prediction methodology, and the five sleep-wake-based seizure prediction approaches.

be problematic if not properly handled [4], [7], [8], [9]. Recognizing and addressing concept drifts could enhance seizure prediction performance, making a real-life implementation more feasible [4], [7], [8], [9], [10], [11]. Specifically, the sleep-wake cycle seems to have a mutual interaction with epilepsy [12], [13], [14], [15], [16], [17], [18], [19]. Seizures often occur during transitional periods of the day, such as the evening before sleep or morning upon waking [4], and also after sleep deprivation [11], [14]. Furthermore, up to half of epileptic patients may experience seizures either only during sleep or only during wakefulness [18], [20]. Seizures exhibit cyclic patterns linked to vigilance states and circadian rhythms [4], [11], [21], [22]. Evidence suggests that seizure susceptibility is more influenced by the vigilance state than by the time of day [11], [16], [18], [20], [21], [22].

Sleep-epilepsy knowledge is limited, and although neglecting sleep-wake information harms seizure prediction algorithm performance [9], [21], few studies address it. Schelter et al. [23] demonstrated that changes in the EEG dynamics associated with the sleep-wake cycle influenced the number of false predictions. Initially, 86% of false alarms were raised during sleep for the dynamic similarity index using a single reference. Motivated by this observation, they combined two reference states (one during sleep and another during wakefulness), which reduced almost 50% of false alarms without significant changes in sensitivity. They also verified that around 20% of false alarms were related to changes in the state of vigilance. In another study, Navarro et al. [24] showed that approximately 80% of preictal dynamic changes were correlated with transitions between states of vigilance. Leal et al. [25] observed a high association between preictal manifestations (obtained through clustering techniques) and the sleep-wake cycle for 5.9% of the seizures with preictal evidence, using univariate linear features.

The states of the sleep-wake cycle fall into the category of reoccurring drifts, which could be handled by having different

classifiers, one for each context [26]. In this case, we refer to two seizure prediction classifiers, each specialized for a different vigilance state (sleep/wakefulness). These states can also be explored as additional features [21]. Based on the aforementioned studies [23], [24], we were also interested in ascertaining the impact of using information about the transitions between vigilance states in seizure prediction.

We implemented five seizure prediction approaches that explicitly consider sleep-wake information. Additionally, we developed a similar seizure prediction pipeline without integrating information about the states of vigilance, which serves as our control method. As determining sleep-wake states is instantaneous, incorporating them into a real-life warning system would be feasible if proven to enhance seizure prediction.

II. MATERIALS AND METHODS

Fig. 1 shows an overview of this study methodology. The first step was developing a supervised sleep-wake classifier, where we used the CAP Sleep database [27]. Next, the sleep-wake model was applied to patients from the EPILEPSIAE database [1], [2] to obtain their vigilance states, which were then incorporated into five patient-specific seizure prediction approaches. We also developed a control method: a seizure prediction pipeline that does not use the sleep-wake model. We developed the sleep-wake model using the CAP Sleep database since the EPILEPSIAE database only provides vigilance state annotations for the last 10 seconds before seizure onset [27]. All code is available for public use at AnaCROliveira.github.

A. Sleep-Wake Detection Methodology

As epilepsy can affect and disturb sleep patterns [14], [15], [16], [17], [18], [19], [28], it may be challenging to obtain accurate sleep-wake classifications with a model trained with healthy subjects or patients with other conditions. For this reason, we

decided to build a sleep-wake detection model based on the literature [29], [30], using only epileptic patients.

1) Algorithm Design: The sleep-wake model was developed using 33 patients diagnosed with nocturnal frontal lobe epilepsy from the CAP Sleep database [27] (available online at PhysioNet [31]). The EEG scalp recordings were pre-processed with a 50 Hz notch filter and a 1-30 Hz bandpass filter. Then, the signals were downsampled to 128 Hz, and 22 univariate linear features were extracted from each 30-second non-overlapping window on 12 EEG bipolar channels.

We randomly divided patients into training and testing sets to avoid these sets containing correlated samples. Therefore, 70% of patients were selected to train an ensemble of 31 linear kernel Support Vector Machine (SVM) classifiers. We have previously implemented a grid-search procedure to find the most suitable number of features and SVM cost hyperparameter.

For testing, the remaining 10 patients' vigilance states were obtained through a majority voting system among the 31 classifiers. The patient-independent model was evaluated with unseen data, demonstrating reliable performance. See Supplementary Section I for more details on the sleep-wake pipeline.

2) Results: The entire methodology was executed 30 times to account for data splitting stochasticity, ensuring statistical confidence in the results. The average testing sensitivity and specificity were 0.93 ± 0.01 and 0.93 ± 0.03 , respectively. Our results are commensurate with the literature [28], [32]. The top-performing sleep-wake model during training achieved testing results of 0.93 for sensitivity and 0.96 for specificity. Refer to Supplementary Section II for further details on the sleep-wake results.

3) Application to EPILEPSIAE Patients: We applied the sleep-wake model to obtain the vigilance states of patients in EPILEPSIAE. Then, we added a post-processing step to smooth the final output by considering its temporality. This step consisted of a sliding-window technique where, for each sample, we assigned the majority class within the previous 10 minutes.

Considering the CAP Sleep testing performance of the best model, reliable classifications were expected. As there were no sleep annotations on the EPILEPSIAE database, we could not directly validate the sleep-wake model's performance. Therefore, we used alternative strategies to validate the results.

The obtained vigilance states were compared to the time of day, considering that most patients sleep at night. Visual inspection was also conducted to analyze artifacts in the EEG signals, including eye blinks, horizontal eye movements, and muscle activity. We also compared our sleep-wake outputs with the EPILEPSIAE annotations provided for the last 10 seconds before seizure onset, obtaining substantial agreement (85%).

Finally, an experienced EEG technician visually inspected and annotated the EEG recordings from an EPILEPSIAE patient (113902). Only one patient was analyzed since manual sleep staging is a time-consuming task. The comparison with our sleep-wake classifications led to a high level of agreement (96.5%). Assuming the manual sleep annotations as ground truth, our model achieved 0.92 for sensitivity and 0.98 for specificity, confirming the successful differentiation of sleep-wake states for external patients.

More details about applying the sleep-wake model to EPILEPSIAE patients and its validation are presented in Supplementary Section III, including the hypnograms comparing sleep scoring and our classifications (Figure S2).

B. Standard Seizure Prediction Methodology

The standard seizure prediction pipeline (Fig. 2) comprised data pre-processing, training, and testing. These steps were individually applied to each patient and are explained in the subsequent sections.

1) Data: For the seizure prediction task, 17 DRE patients (8 females and 9 males; age range: 13-65 years; mean age: 32 ± 16 years) were selected from the European Epilepsy Database [1], [2]. We analyzed data recorded during presurgical monitoring at the Epilepsy Center of the University Hospital of Freiburg in Germany.

The remaining selection criteria were the following: i) patients containing a minimum of 4 seizures, with at least 4.5 hours between each seizure, ii) patients having at least one seizure from each vigilance state (sleep/wakefulness) within the first three seizures, and iii) patients with EEG scalp recordings acquired with a sampling frequency of 256 Hz. The signals cover 19 EEG electrodes placed according to the International 10-20 System, organized in a referential montage.

Considering the set of patients used to develop the sleep-wake classifier, it would be ideal to choose patients with frontal lobe seizures for this step. However, only two patients met all the criteria, which led to the inclusion of patients with temporal lobe seizures, as these are the most representative among DRE patients [12]. Applying these criteria resulted in a dataset of 94 seizures (51 for training, 43 for testing) and a total of 1970 hours of data (1488 hours for training, 482 hours for testing). More details about the dataset can be found in Table S3, available in Supplementary Section IV.

The use of data from EPILEPSIAE for research purposes has been authorized by the Ethical Committee of the three hospitals involved in the database development (Ethik-Kommission der Albert-Ludwigs-Universität, Freiburg; Comité consultatif sur le traitement de l'information en matière de recherche dans le domaine de la santé, Pitié-Salpêtrière University Hospital; and Comité de Ética do Centro Hospitalar e Universitário de Coimbra). All methods were performed following the relevant guidelines and regulations. Informed written consent was obtained from all patients.

2) Pre-Processing and Feature Extraction: The EEG recordings were pre-processed using a deep convolutional neural network-based model for artifact removal [33], which mimicked expert manual pre-processing. This model was developed with a subset of Freiburg patients from the EPILEPSIAE database.

Since the signals were acquired using a referential montage, we re-referenced them to obtain the following EEG bipolar traces [34]: FP2-F4, F4-C4, C4-P4, P4-O2, FP1-F3, F3-C3, C3-P3, P3-O1, F8-T4, T4-T6, F7-T3, and T3-T5. Then, the data was bandpass filtered from 1-30 Hz and downsampled from 256 Hz to 128 Hz. We did these steps to use the signals within the same conditions as the ones from the CAP Sleep database.

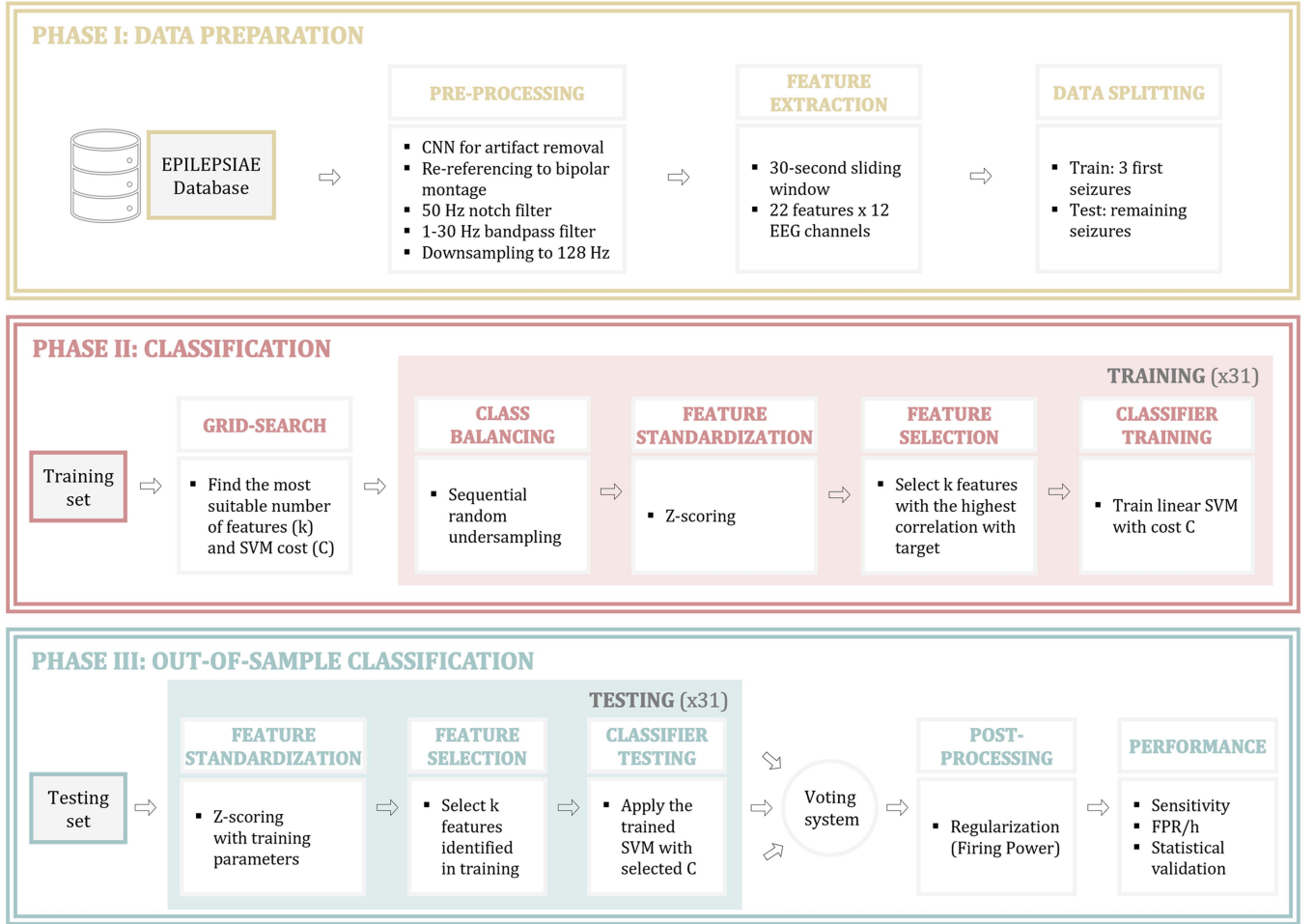


Fig. 2. Flowchart of the standard seizure prediction methodology applied to each patient, including data preparation, classification, and out-of-sample classification.

The EEG signals were segmented into non-overlapping 30-second windows. For each time window and EEG channel, 22 univariate linear features were extracted: i) statistical moments (mean, variance, skewness, kurtosis); ii) relative spectral power in delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–25 Hz); iii) spectral edge power and spectral edge frequency (50%, 75%, 90%); iv) Hjörth parameters (mobility, complexity); and v) energy of wavelet coefficients at five decomposition levels (detail coefficients D1-D5, approximation coefficient A5). We selected these features for their simplicity, quick computation, and wide adoption in seizure prediction research [6], [9], [35], [36], [37], [38], [39].

3) Classification: A patient-specific approach was employed for seizure prediction, where the first three seizures were used for training and the remaining for testing.

Two classes were considered: interictal and preictal. Preictal labels were assigned to windows prior to seizure onset according to the duration of SOP and SPH. We set the SPH to 5 minutes [40] while considering a scalp EEG warning system. This system could allow patients to minimize seizure consequences, either by preparing for the upcoming seizure (e.g., stopping driving a car) or taking rescue medication, such as benzodiazepines (e.g., a diazepam nasal spray, FDA-approved in 2020, requires less

than 5 minutes to take effect [41]). The SOP value ranged from 10 to 60 minutes in 5-minute increments, which we considered reasonable according to patients' preferences [42].

To address class imbalance during training, we implemented a random undersampling with a chronological constraint to maintain data representativeness. Interictal samples were divided into n chronological groups, and one sample was randomly selected from each group while discarding the rest. The n denotes the number of preictal samples.

We employed the Analysis of Variance (ANOVA) f -test to select the top k most discriminative features. This filter method performs a feature ranking according to the degree of linear dependency between each feature and the target classes [43]. The SVM classifier with a linear kernel was chosen for seizure prediction due to its simplicity and comparable performance [35].

A grid-search procedure was conducted to determine the optimal number of features (k) and to tune the SVM hyperparameter (C). The search explored 5 values of k (10, 20, 30, 40, 50) and 17 values of C (2^{-20} , 2^{-18} , ..., 2^{12}), resulting in 85 combinations. To ascertain the best combination (k , C), we applied a 3-fold cross-validation technique, where each fold concerned data from one of the training seizures, implying that each seizure was used as the validation set exactly once. In each iteration, the training

set (2 seizures) was balanced, standardized, and used to find the best set of k features to train the SVM classifier considering C . Since the validation set (1 seizure) was not balanced, the performance metric was based on sample sensitivity (SS) and specificity (SP) as follows: $\sqrt{SS \times SP}$ [44]. Considering the undersampling stochasticity, each iteration was performed 31 times. In sum, the final metric of each combination (k , C) was obtained by averaging performances over the 3 iterations and 31 classifiers within each iteration. After evaluating all the combinations (k , C), the one with the highest metric yielded the optimal parameters.

We then trained 31 classifiers using the tuned parameters (k , C) and the entire set of training data (3 seizures). Several classifiers were trained due to the randomness of the class balancing procedure. The rationale behind the number 31 was to achieve statistical confidence in the results [45] and avoid ties in the testing phase.

4) Out-of-Sample Classification: In the testing phase, we used the remaining seizures of each patient to perform out-of-sample classification. As shown in Fig. 2, the same procedure of the training step was applied to the testing set, excluding the class balancing process, as it would not represent a real-life implementation.

When applying the trained seizure prediction model to new data, the following steps are undertaken: i) standardization of the new data using the z-score parameters from the training data, ii) selection of the most discriminative features identified during training from the testing dataset, and iii) application of the trained SVM classifier.

As 31 classifiers were trained, this procedure was repeated for each one, resulting in 31 output predictions for each sample. A majority voting system determined the final output, where an odd number of classifiers avoided ties. In this case, the predominant class corresponds to the one comprised of at least 16 predictions.

5) Post-Processing: After classification, we implemented a regularization method to smooth the final output by considering its temporal dynamics [6]. We chose the Firing Power [46], [47], a sliding window technique that is conservative in the generation of alarms. The Firing Power $fp[n]$, at instant n , was calculated using (1), where τ represents the number of samples within a window of time with the preictal duration, and $O[k]$ is the binary output of the classifier for each sample k . The resulting measure, ranging from 0 to 1, indicates the proportion of samples classified as preictal within the sliding window.

$$fp[n] = \frac{\sum_{k=n-\tau}^n O[k]}{\tau} \quad (1)$$

The Firing Power was first computed for each window. Then, an alarm was raised when it simultaneously exceeded the established threshold of 0.5 and was at least one refractory period apart from the last generated alarm. The refractory period was defined as the total time during which the patient is preparing for the seizure (SPH) and expecting its occurrence (SOP). This constraint reduces multiple consecutive alarms, minimizing the patient's confusion and stress in real-life implementation.

6) Performance Evaluation: We evaluated performance with the standard metrics: seizure sensitivity (SS) and false

positive rate per hour (FPR/h) [5], [48], [49]. SS is defined by (2) and depicts the proportion of correctly predicted seizures.

$$SS = \frac{N_{true\ alarms}}{N_{seizures}} \quad (2)$$

FPR/h measures the false alarm rate. This metric might be helpful to explore the most suitable prediction applicability according to the patient's resilience towards false interventions. FPR/h is defined by (3), where the denominator refers to the total time during which false alarms can be raised. Therefore, the refractory period after each alarm is subtracted from the interictal period due to the impossibility of triggering new alarms during these periods.

$$FPR/h = \frac{N_{false\ alarms}}{Interictal - N_{false\ alarms} \times (SOP + SPH)} \quad (3)$$

Besides the assessment of these performance metrics, statistical validation is a core aspect in the field of seizure prediction. The seizure-times surrogates method [50], [51] was implemented to verify if the proposed algorithm performs better than chance. This method replaces the original seizure onset times with artificial onsets randomly selected from the interictal period. The surrogate sensitivity is computed using these new labels. This process is repeated 30 times, and the average surrogate sensitivity is compared to the sensitivity of the methodology under study. The proposed methodology performs above chance if its sensitivity is higher than the surrogate sensitivity with a statistical significance of $\alpha=0.05$, determined through a one-sample t-test. The null hypothesis is that the sensitivity of the proposed methodology is not better than the surrogate sensitivity.

C. Sleep-Wake-Based Seizure Prediction Methodologies

Taking into account circadian information regarding vigilance states (sleep/wakefulness), we propose five patient-specific approaches for seizure prediction. These approaches require modifications to the standard seizure prediction methodology (Control). This section provides further details on these alternative approaches.

1) Feature_{state} Approach: The Feature_{state} approach aims to improve seizure prediction by considering physiologically relevant modulators of epileptic activity, such as the sleep-wake state. Based on recommendations [21] and inspired by a previous study [52], we included vigilance states as an additional feature of the seizure prediction model. This inclusion was mandatory, regardless of the feature selection process, which only needed to choose the $k - 1$ most discriminative features.

2) Pool_{weights} Approach: The Pool_{weights} approach involves the creation of two separate seizure prediction models: one for sleep and one for wakefulness. Each model was trained using a weighted SVM classifier, where different weights were assigned to samples based on the vigilance state. For instance, regarding the implementation of the sleep model, sleep samples had a higher weight than awake samples. Consequently, during the train, the SVM classifier assigned more importance to sleep samples in regard to discerning between preictal and interictal.

TABLE I
OVERVIEW AND DESCRIPTION OF ALL SEIZURE PREDICTION APPROACHES

Approach	Description	Train	Post-processing method
Control	Standard seizure prediction algorithm, the control method, that does not use sleep-wake information.	1 model	1 threshold: · 0.5 for all samples
Feature _{state}	Seizure prediction algorithm with an extra feature of the vigilance state in model training.	1 model with additional feature	1 threshold: · 0.5 for all samples
Pool _{weights}	Seizure prediction algorithm with a pool of two distinct models, each trained with all samples according to weights , assigned based on the vigilance state.	2 models: · sleep model with sleep samples (weight 0.8) and awake samples (weight 0.2) · awake model with awake samples (weight 0.8) and sleep samples (weight 0.2)	1 threshold: · 0.5 for all samples
Pool _{exclusive}	Seizure prediction algorithm with a pool of two distinct models, each trained with samples exclusively from the respective vigilance state.	2 models: · sleep model only with sleep samples · awake model only with awake samples	1 threshold: · 0.5 for all samples
Threshold _{state}	Seizure prediction algorithm with a different threshold for each vigilance state in the post-processing phase.	1 model	2 thresholds: · 0.5 for sleep samples · 0.75 for awake samples
Threshold _{transitions}	Seizure prediction algorithm with a different threshold for vigilance state transitions in the post-processing phase.	1 model	2 thresholds: · 0.25 for vigilance transitions · 0.5 for the remaining samples

It was reasonably decided that the weakest weight would be a quarter of the strongest weight. We have not considered more discrepant weights to avoid a possible confounding effect on the discernment of preictal and interictal samples. Table I presents the configuration of each model.

The motivation behind different training weights (and not deleting samples) is the relatively scarce amount of training samples from both vigilance states, especially regarding the preictal period. Also, including, for instance, sleep samples into the awake model can be considered reasonable due to the existence of psychological states that occur during both vigilance states [13], [48].

Each model had its own grid-search procedure to tune specific parameters as in the standard methodology. During the testing phase, one of the trained models from the pool of predictors was selected according to the vigilance state of each window.

3) Pool_{exclusive} Approach: The Pool_{exclusive} approach also involves creating separate sleep and awake models as a pool of predictors. Unlike the Pool_{weights} approach, each model was exclusively trained with samples from its respective vigilance state. Some patients (202, 12702, 30802, 50802, 102202) were excluded from this approach because they did not have both sleep and awake samples during the preictal periods. Despite our patient selection criteria, the discrepancy between our sleep-wake classifier and the EPILEPSIAE vigilance state annotation during seizure onset could be explained by: i) wrong predictions of the sleep-wake classifier, ii) incorrect manual sleep annotations, or iii) a patient waking up in the last minutes before a seizure, where we used a 5-minute SPH. Henceforth, Control* refers to the standard methodology using only the patients included in this approach, thereby ensuring a fair comparison.

We modified the grid-search procedure to accommodate the absence of both sleep and awake samples in the preictal period of each seizure. Instead of the previous 3-fold cross-validation, with each fold containing samples from a single seizure, we implemented a traditional 3-fold cross-validation by randomly splitting all samples from the three seizures based on the proportion of preictal and interictal samples. Similar to the Pool_{weights}

approach, each model underwent its own grid-search procedure, and during the testing phase, the model to be applied was selected based on the vigilance state.

4) Threshold_{state} Approach: The Threshold_{state} approach aims to incorporate particular considerations for different contexts, specifically the sleep-wake states. In this approach, the regularization phase takes into account the vigilance states by employing two thresholds, instead of just one, for the Firing Power. The sleep threshold remains at the default value of 0.5, whereas the wakefulness threshold is set at 0.75. These thresholds were reasonably defined according to the following assumptions.

Sleep is characterized by reduced mobility, consciousness, and responsiveness compared to wakefulness [53], which, in turn, involves daily-life activities that introduce more artifacts in the EEG. Thus, a higher threshold during wakefulness periods may prompt the classifier to be more robust to these artifacts, reducing the number of false alarms. Also, since seizures are more common during sleep [54], a lower threshold is more suitable for capturing preictal changes during this seizure-prone period. Finally, Sudden Unexpected Death in Epilepsy (SUDEP) occurs more frequently during sleep [13], [21]. Therefore, employing a less conservative threshold that may trigger more true alarms could be seen as a preventive measure to promptly alert caregivers and potentially mitigate such tragedies.

5) Threshold_{transitions} Approach: The Threshold_{transitions} approach is driven by the increased seizure susceptibility during transitions between vigilance states [24], [48]. Patients often report a propensity for seizures during sleep/awake or awake/sleep transitions [4]. Furthermore, a significant portion of false alarms is attributed to changes in vigilance state [23].

Based on this evidence, this approach incorporates the vigilance states into the regularization phase. Once again, two thresholds are used for the Firing Power. A transitional threshold of 0.25 was reasonably defined and applied for 30 minutes after transitioning between awake and sleep states. For other periods, the default threshold of 0.5 was used. By recognizing that vigilance fluctuations amplify seizure susceptibility, adopting

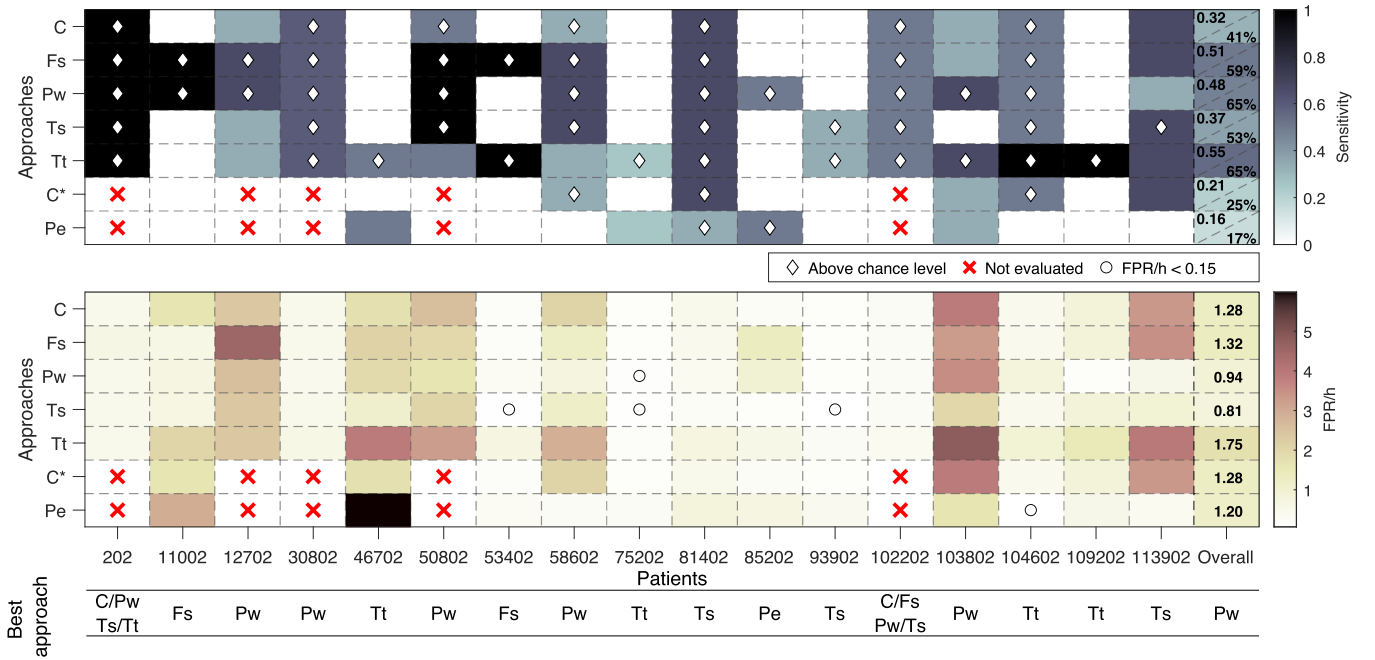


Fig. 3. Test performance of all patients and approaches: C (Control), Fs (Feature_{state}), Pw (Pool_{weights}), Ts (Threshold_{state}), Tt (Threshold_{transitions}), C* (Control*), and Pe (Pool_{exclusive}). On the top, the color scale represents the seizure sensitivity, and the diamond shape is present when the patient is statistically validated. For each approach, the overall column contains the average seizure sensitivity and the percentage of patients whose models performed above the chance level. On the bottom, the colors represent the false prediction rate per hour, and the circle shape is present if its value is considered adequate (<0.15 [5]). The overall column contains the average FPR/h for each approach. The red cross refers to patients that were not evaluated.

a lower transitional threshold enhances the detection of preictal evidence.

III. RESULTS AND DISCUSSION

Fig. 3 presents an overview of the test performance for each patient and all approaches, including sensitivity, FPR/h, and statistical validation.

Only 24% (4 in 17) of the patients performed above the chance level in all tested approaches. However, all patients (17 in 17) were statistically validated by at least one approach, implying the need for personalized strategies. In contrast to the control method, which only validated 41% (7 in 17) of the patient models, all 17 patients achieved significant results for at least one of the seizure prediction methods that incorporated the vigilance states. While no approach was consensual among all patients, the Pool_{weights} and Threshold_{transitions} approaches were the ones that validated the highest amount of patients (11 out of 17). Among patients that achieved significant results with the control, all performed above chance with the Pool_{weights} approach, whereas the Threshold_{transitions} approach failed to validate two (50802 and 58602).

In general, we observed two prevalent scenarios in the test results: 7 out of 17 patients performed better than chance with the control and most sleep-wake-based approaches, while the remaining 10 were only statistically validated by one or two of the methods that integrated the vigilance states. The latter might represent more complex cases where it may be harder to capture pre-seizure generation mechanisms. Incorporating vigilance states has proven beneficial in these cases.

Additionally, apart from the sleep-wake information, we hypothesize that further concept drifts could assist in distinguishing preictal and interictal periods.

Overall, except for the Pool_{exclusive} approach, the remaining sleep-wake-based approaches outperformed the control in terms of sensitivity. Although the Threshold_{transitions} approach showed the highest improvement, the correspondent FPR/h increased. Undoubtedly, both performance metrics must be evaluated simultaneously since an excessive number of false alarms may prompt adverse consequences, such as leading to unnecessary interventions that could be harmful and worsen a situation of helplessness. In the case of a warning system, it could lead patients to discredit the system and consequently ignore warnings. In presurgical monitoring, where seizures are more frequent due to sleep deprivation and medication withdrawal [4], [14], [15], [16], a maximum FPR/h value of 0.15 is considered reasonable regarding a real-life warning system [5]. As depicted in Fig. 3, this condition was rarely fulfilled. In fact, only one patient (93902) was statistically validated while presenting an adequate FPR/h value (<0.15 [5]), and this solely occurred with the Threshold_{state} approach.

Globally, compared to the control, the Pool_{weights} and Threshold_{state} approaches were the only ones that improved all evaluation aspects. Nonetheless, the Feature_{state} approach also increased the percentage of validated patients and sensitivity, with a minor increase in the average FPR/h value. Overall, the Pool_{weights} approach presented the highest performance increase. This approach trained two seizure prediction models, each specialized in a different vigilance state according to weights.

TABLE II

AVERAGE TEST RESULTS OF THE SEIZURE PREDICTION MODELS FOR ALL APPROACHES: C (CONTROL), Fs (FEATURE_{STATE}), Pw (POOL_{WEIGHTS}), Ts (THRESHOLD_{STATE}), Tt (THRESHOLD_{TRANSITIONS}), C* (CONTROL*), AND PE (POOL_{EXCLUSIVE})

Approach	#	All patients				Validated patients				Sleep		Awake	
		SOP (min)	SS (0-1)	FPR/h	%	SOP (min)	SS (0-1)	FPR/h	SS (0-1)	# False alarms	SS (0-1)	# False alarms	
C	17	33 ± 19	0.32 ± 0.31	1.28 ± 1.18	41	40 ± 21	0.59 ± 0.19	1.00 ± 0.87	0.41	193 (61%)	0.19	121 (39%)	
Fs	17	35 ± 19	0.51 ± 0.38	1.32 ± 1.29	59	37 ± 19	0.76 ± 0.20	1.09 ± 1.25	0.56	226 (69%)	0.31	103 (31%)	
Pw	17	34 ± 19	0.48 ± 0.35	0.94 ± 0.92	65	32 ± 20	0.71 ± 0.19	1.15 ± 0.99	0.52	179 (66%)	0.38	94 (34%)	
Ts	17	33 ± 19	0.37 ± 0.35	0.81 ± 0.71	53	43 ± 19	0.66 ± 0.21	0.70 ± 0.59	0.52	203 (79%)	0.19	53 (21%)	
Tt	17	33 ± 19	0.55 ± 0.32	1.75 ± 1.47	65	40 ± 20	0.68 ± 0.27	1.33 ± 1.48	0.48	219 (59%)	0.56	150 (41%)	
C*	12	30 ± 19	0.21 ± 0.26	1.28 ± 1.24	25	32 ± 8	0.50 ± 0.14	1.02 ± 0.77	0.31	44 (29%)	0.20	108 (71%)	
Pe	12	20 ± 14	0.16 ± 0.20	1.20 ± 1.63	17	17 ± 8	0.42 ± 0.08	0.75 ± 0.05	0.15	45 (31%)	0.20	101 (69%)	

It includes the SOP, seizure sensitivity, and FPR/h values, both for all patients and for those who performed above chance level, followed by the seizure sensitivity and number of false alarms obtained for each vigilance state.

Nevertheless, not even the pool of weighted predictors approach was consensual among all patients. Different cases appear to vary in complexity, indicating the need for tailored strategies depending on the individual, as previously observed. We selected the best method for each case based on statistical validation, sensitivity, and FPR/h. As demonstrated in Fig. 3, the Pool_{weights} approach was the most chosen method, with 7 selections, followed by the Threshold_{state} and Threshold_{transitions} approaches (5 selections each). On the other hand, the Pool_{exclusive}, Feature_{state}, and Control approaches had fewer preferences, chosen only 1, 2, and 3 times, respectively. Since no clear patterns emerged among the patients, it is challenging to ascertain the best approach for each case beforehand. However, if this study were used as a preliminary test to determine subsequent method selection for each patient, the Pool_{weights} approach would be chosen 41% of the time.

Table II presents the average results for all approaches. Sensitivity values exhibit large standard deviations, indicating significant performance variations among patients. On the other hand, we only tested an average of 2.53 seizures per patient. For instance, computing the sensitivity when considering only one testing seizure restricts its value either to 0 (missed seizure) or 1 (predicted seizure). As almost half of our patients have less than three seizures for testing, this limitation contributes to such high standard deviations. To address this, we employed a surrogate predictor. Although computationally heavier than a random predictor, it is more flexible and robust, providing greater confidence in the results [8], [50].

Table II demonstrates that the sleep-wake-based approaches, except for Pool_{exclusive}, outperformed the control in terms of sensitivity and statistical validation percentage. Only the Pool_{weights} and Threshold_{state} approaches reduced the FPR/h as well. Considering all factors, the Pool_{weights} approach, the pool of weighted predictors, was indeed our best method. Approximately 65% of the patients performed better than chance with this proposal, whereas the control statistically validated less than half (41%). Furthermore, the control presents average sensitivity and FPR/h values of 0.32 ± 0.31 and 1.28 ± 1.18 , respectively. In turn, our best method achieved 0.48 ± 0.35 for sensitivity and 0.94 ± 0.92 for FPR/h. Despite the improvement, these results lack clinical validation. Even when considering the average results for the statistically validated patients, the

sensitivity values are substantially better, but the FPR/h values are still unacceptable regarding a real-life warning system. A neurostimulation closed-loop system could be a reasonable alternative since there may be a higher tolerance for false alarm interventions [55].

We selected the SOP duration for each patient based on the remaining training metrics. Table II displays an average SOP value of approximately 30 minutes for all approaches except Pool_{exclusive} (20 minutes). These values exhibit large standard deviations, with selected SOPs ranging from 10 to 60 minutes. This suggests that the optimal preictal duration may vary among patients, highlighting the challenge of establishing a standard duration. Considering the validated individuals among the 17 patients, the Pool_{weights} approach has the shortest mean SOP duration, providing an additional advantage by possibly minimizing stress and anxiety.

Among our approaches, we have two similar conceptualizations: Pool_{weights} and Pool_{exclusive}. These approaches train two seizure prediction models, each specialized in a different vigilance state. However, the results from these approaches were unexpectedly disparate. Potential explanations are discussed below.

Firstly, the average performance metrics and statistical validation rate of the Pool_{exclusive} approach are based on a smaller patient cohort. Secondly, we implemented different cross-validation strategies in the grid-search procedure of each approach, which highly influenced the chosen parameters (number of features and SVM cost) and, consequently, the selected SOPs. In fact, the average SOP durations were 34 ± 19 and 20 ± 14 minutes for the Pool_{weights} and Pool_{exclusive} approaches, respectively. Finally, the Pool_{weights} approach trains models using all samples weighted by vigilance state, whereas the Pool_{exclusive} approach exclusively uses samples from the respective vigilance state. This restriction of training with samples from a single sleep stage may be overly limiting, as psychological states can occur during both vigilance states.

Besides wakefulness and sleep, different sleep stages exhibit electrophysiological distinctions that affect the origin, suppression, and propagation of seizures [15]. In fact, Rapid Eye Movement (REM) sleep has been found to have antiepileptic effects, with most seizures occurring during Non-Rapid Eye Movement (NREM) stages and wakefulness [13], [14], [15], [16], [17], [18],

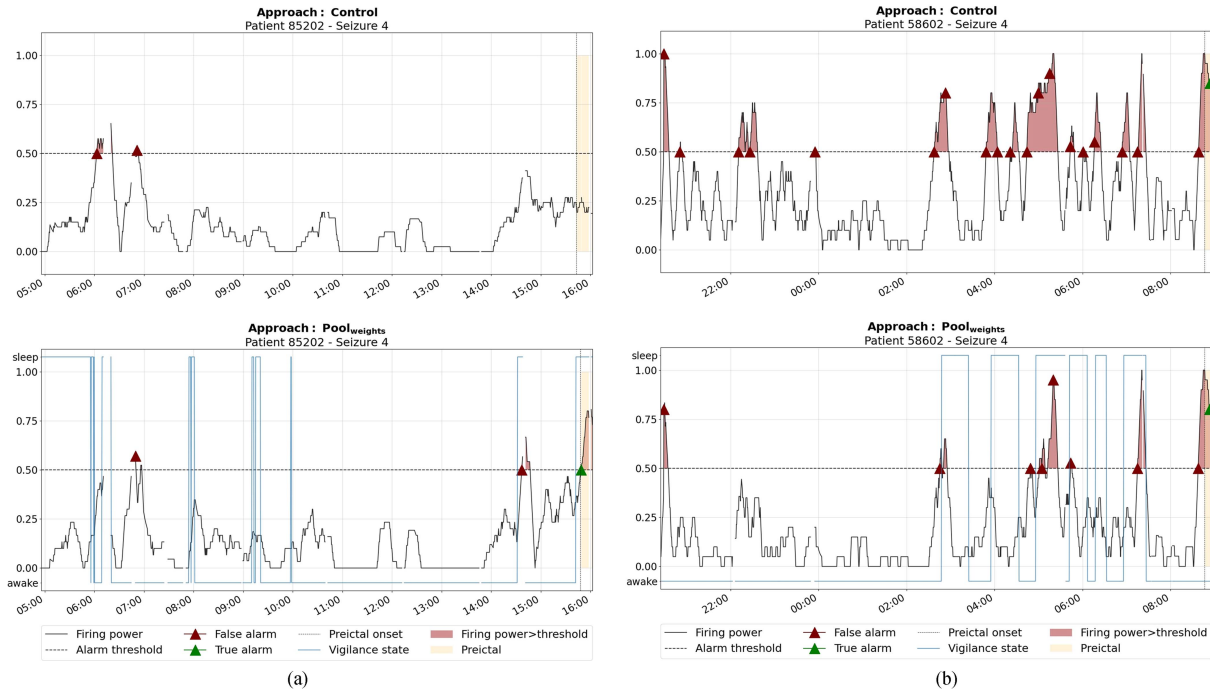


Fig. 4. Plots of the firing power over time, along with the sleep/awake state and raised alarms. (a) Example showing that the control was not able to predict the seizure, whereas the Pool_{weights} approach predicted it. (b) Example showing that the control raised 19 false alarms, whereas the Pool_{weights} approach raised 8 false alarms.

[21], [54]. Furthermore, bed and wake times seem to influence seizure risk [16]. It would be of interest to explore these aspects in future treatments, either by integrating such information into seizure prediction approaches or by applying it in chronotherapy or presurgical investigations.

Regarding the Threshold_{state} and Threshold_{transitions} approaches, we could have tested several thresholds. However, we have not executed performance optimizations based on the testing set to avoid publication bias. In Supplementary Section V, we only present the results that one would reach with assumptions contrary to ours.

As for the Feature_{state} approach, we hypothesize whether neural networks with sufficient complexity could intrinsically handle and incorporate the sleep-wake concepts into seizure prediction. However, this cannot be guaranteed, which led us to employ a more interpretable approach.

Table II also presents the seizure sensitivity and number of false alarms for each vigilance state. Across the 17 patients, there is a tendency for a higher percentage of false alarms during sleep, consistent with findings by Schelter et al. [23]. It is worth mentioning, however, that 59% of our testing data refers to sleeping periods. Furthermore, the Control* and Pool_{exclusive} approaches, which only included 12 patients and 43% of sleeping data, show more wrong predictions when they were awake. Therefore, we believe no general conclusion can be drawn, as it seems to depend on the set of patients under analysis. For instance, regarding the control method, five patients only had false alarms during sleep, five during wakefulness, and the remaining seven during both states. In contrast, considering the same method for the set of 12 patients, only two had incorrect

predictions solely during sleep and five during wakefulness periods, thus explaining the discrepancy in the results.

Compared to the control, only the Pool_{weights} improved the sensitivity and reduced the number of false alarms regarding each vigilance state. Specifically, our best method predicted 38% of wakefulness seizures and 52% of sleep seizures, even though around 66% of false alarms occurred during sleep. Regarding the Threshold_{state} approach, we applied a higher threshold of 0.75 for wakefulness samples in the post-processing phase. Since it is harder to cross the adapted threshold, even in the case of an emerging seizure, we expected to achieve a lower number of false alarms and possibly a lower sensitivity value for awake periods. Nevertheless, we were able to predict the same amount of wakefulness seizures with a reduction of 56% of false alarms during this state. This approach also correctly predicted more sleep seizures than the control, even though incorrect predictions increased by 5%. As for the Threshold_{transitions} approach, a lower threshold of 0.25 was implemented during sleep-wake transitions. This approach achieved higher sensitivity at the cost of additional incorrect predictions for both vigilance states.

Next, we visually inspected the time plots of seizure probability occurrence to gain insights into the models' behavior beyond performance metrics [39]. Fig. 4(a) illustrates an example where the control failed to predict the seizure, whereas our best approach successfully predicted it. In another example (Fig. 4(b)), both approaches predicted the seizure, but the Pool_{weights} approach raised less than half of the false alarms.

After analyzing our results, we can compare them with previous studies that used patients from the EPILEPSIAE database to develop seizure prediction models [36], [37], [38], [39], [49].

Other studies [35], [56] cannot be compared as the authors trained and tested several models, reporting only the best testing results. Although these results provide a more optimistic look, they are overestimated and do not reflect a real-time implementation. Equivalently, if we had selected the SOP values based on the testing performance, our best method would have achieved 0.75 ± 0.24 for sensitivity and 0.62 ± 0.55 for FPR/h, with all patients performing better than chance.

Alvarado-Rojas et al. [49] achieved a sensitivity of 0.47 and FPR/h of 0.94, which aligns with our best method. They have not considered SPH, i.e., any time reserved to revert or prepare for the upcoming seizure. In contrast, we defined an SPH value of 5 minutes to account for warning systems followed by rescue medication intake. It is worth noting that we could have benefited from analyzing data comprised 5 minutes before seizure onsets. As it is closer to the event, it has a higher likelihood of containing preictal evidence. Despite only statistically validating 13% of the patients (7 in 53), which is significantly lower than the rate of validated patients by our control (41%) and by our best approach (65%), they used a random predictor instead of a surrogate analysis.

Pinto et al. [36], [37] employed surrogate analysis, obtaining significant results for 32% of patients. Both studies used evolutionary algorithms and defined an SPH of 10 minutes. Lopes et al. [38] obtained an average sensitivity of 0.37 ± 0.36 and FPR/h of 0.86 ± 0.77 with denoised features and a shallow neural network. They also employed surrogate analysis, achieving performance above the chance level for 54% (22 in 41) of the patients. These studies [36], [37], [38] applied a chronological approach to address concept drifts over time by retraining the models after every new seizure. Although using all available interictal data is highly recommended [9], they only used the last 4 hours before each seizure to train the prediction models due to computational reasons. Since the preictal state can manifest minutes to hours before seizure onset [7], [9], training with a 4-hour window may lack sufficient representation of the interictal state, thus limiting its discrimination from the preictal state.

Pinto et al. [39] used all the available interictal data, achieving 0.17 ± 0.28 for sensitivity and 0.87 ± 1.11 for FPR/h with an ensemble of 15 SVMs. They statistically validated 17.5% of patients using a surrogate analysis.

IV. CONCLUSION

This work aimed to incorporate sleep-wake information into EEG epilepsy seizure prediction algorithms. We developed a sleep-wake detection model that achieved satisfactory results (0.93 ± 0.01 for sensitivity, 0.93 ± 0.03 for specificity). Since the CAP Sleep database only comprises night recordings, we believe these results could be further improved with a database containing both day and night recordings. Except for one approach, the patient-specific proposals that considered the vigilance states outperformed the control. Although our findings reveal the need for different strategies concerning different patients, the most widely accepted method was the Pool_{weights} approach, which

attained an average seizure sensitivity of 0.48 ± 0.35 , an average FPR/h of 0.94 ± 0.92 , and 65% of patients performing above chance level. For future guidance, this approach could be improved with patient-tailored weights, optimized during cross-validation.

We emphasize that the testing performance was obtained with unseen data, where all the parameters were previously optimized with the training set (pseudo-prospective procedure). Additionally, our choices reasonably contemplated the ultimate goal of prospectively running the algorithms in a real-life scenario. Nevertheless, the major limitation of the present study is related to the data, which was acquired during presurgical monitoring. The extent to which sleep deprivation and medication withdrawal have affected our findings remains uncertain. Consequently, it is necessary to reproduce this study with long-term data collected from daily-life conditions to truly assess its future applicability.

Besides sleep and wakefulness, we hypothesize that incorporating further sleep stage differentiation into novel seizure prediction approaches could enhance performance. In addition to the sleep-wake cycle, other concept drifts may also aid in distinguishing preictal and interictal periods. For example, periodic retraining of seizure prediction models could be explored [26]. This study contributes to the seizure prediction field and guides the design of new approaches, offering opportunities to enhance the performance of seizure prediction algorithms.

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